

Foundations of Deep Q-network

[Deep Q-network]

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simple cells, whose output is maximized by straight edges having particular orientations within their receptive field complex cells, which have larger receptive fields, whose output is insensitive to the exact position of the edges in the field. Hubel and Wiesel also proposed a cascading model of these two types of cells for use in pattern recognition tasks.

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Evaluation The accuracy of the final model is typically estimated on a sub-part of the dataset set apart at the start, often called a test set. Alternatively, methods such as k-fold cross-validation are applied. Other strategies include using conformal prediction.

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Filter (or kernel) size Common filter sizes found in the literature vary greatly, and are usually chosen based on the data set. Typical filter sizes range from 1x1 to 7x7. As two famous examples, AlexNet used 3x3, 5x5, and 11x11. Inceptionv3 used 1x1, 3x3, and 5x5. The challenge is to find the right level of granularity so as to create abstractions at the proper scale, given a particular data set, and without overfitting.

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Natural language processing CNNs have also been explored for natural language processing. CNN models are effective for various NLP problems and achieved excellent results in semantic parsing, search query retrieval, sentence modeling, classification, prediction and other traditional NLP tasks. Compared to traditional language processing methods such as recurrent neural networks, CNNs can represent different contextual realities of language that do not rely on a series-sequence assumption, while RNNs are better suitable when classical time series modeling is required.

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For example, in CIFAR-10, images are only of size 32×32×3 (32 wide, 32 high, 3 color channels), so a single fully connected neuron in the first hidden layer of a regular neural network would have $32 \times 32 \times 3 = 3,072$ weights. A 200×200 image, however, would lead to neurons that have $200 \times 200 \times 3 = 120,000$ weights. Also, such network architecture does not take into account the spatial structure of data, treating input pixels which are far apart in the same way as pixels that are close together. This ignores locality of reference in data with a grid-topology (such as images), both computationally and semantically. Thus, full connectivity of neurons is wasteful for purposes such as image recognition that are dominated by spatially local input patterns. Convolutional neural networks are variants of multilayer perceptrons, designed to emulate the behavior of a visual cortex. These models mitigate the challenges posed by the MLP architecture by exploiting the strong spatially local correlation present in natural images. As opposed to MLPs, CNNs have the following distinguishing features:

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Notable libraries Caffe: A library for convolutional neural networks. Created by the Berkeley Vision and Learning Center (BVLC). It supports both CPU and GPU. Developed in C++, and has Python and MATLAB wrappers. Deeplearning4j: Deep learning in Java and Scala on multi-GPU-enabled Spark. A general-purpose deep learning library for the JVM production stack running on a C++ scientific computing engine. Allows the creation of custom layers. Integrates with Hadoop and Kafka. Dlib: A toolkit for making

real world machine learning and data analysis applications in C++. Microsoft Cognitive Toolkit: A deep learning toolkit written by Microsoft with several unique features enhancing scalability over multiple nodes. It supports full-fledged interfaces for training in C++ and Python and with additional support for model inference in C# and Java. TensorFlow: Apache 2.0-licensed Theano-like library with support for CPU, GPU, Google's proprietary tensor processing unit (TPU), and mobile devices. Theano: The reference deep-learning library for Python with an API largely compatible with the popular NumPy library. Allows user to write symbolic mathematical expressions, then automatically generates their derivatives, saving the user from having to code gradients or backpropagation. These symbolic expressions are automatically compiled to CUDA code for a fast, on-the-GPU implementation. Torch: A scientific computing framework with wide support for machine learning algorithms, written in C and Lua.

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LeNet-5, a pioneering 7-level convolutional network by LeCun et al. in 1995, classifies hand-written numbers on checks digitized in 32×32 pixel images. The ability to process higher-resolution images requires larger and more layers of convolutional neural networks, so this technique is constrained by the availability of computing resources. It was superior than other commercial courtesy amount reading systems (as of 1995). The system was integrated in NCR's check reading systems, and fielded in several American banks since June 1996, reading millions of checks per day.

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Time series forecasting Recurrent neural networks are generally considered the best neural network architectures for time series forecasting (and sequence modeling in general), but recent studies show that convolutional networks can perform comparably or even better. Dilated convolutions might enable one-dimensional convolutional neural networks to effectively learn time series dependences. Convolutions can be implemented more efficiently than RNN-based solutions, and they do not suffer from vanishing (or exploding) gradients. Convolutional networks can provide an improved forecasting performance when there are multiple similar time series to learn from. CNNs can also be applied to further tasks in time series analysis (e.g., time series classification or quantile forecasting).

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Time delay neural networks The time delay neural network (TDNN) was introduced in 1987 by Alex Waibel et al. for phoneme recognition and was an early convolutional network exhibiting shift-invariance. A TDNN is a 1-D convolutional neural net where the convolution is performed along the time axis of the data. It is the first CNN utilizing weight sharing in combination with a training by gradient descent, using backpropagation. Thus, while also using a pyramidal structure as in the neocognitron, it performed a global optimization of the weights instead of a local one. TDNNs are convolutional networks that share weights along the temporal dimension. They allow speech signals to be processed time-invariantly. In 1990 Hampshire and Waibel introduced a variant that performs a two-dimensional convolution. Since these TDNNs operated on spectrograms, the resulting phoneme recognition system was invariant to both time and frequency shifts, as with images processed by a neocognitron. TDNNs improved the performance of far-distance speech recognition.

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"S-layer": a shared-weights receptive-field layer, later known as a convolutional layer, which contains units whose receptive fields cover a patch of the previous layer. A shared-weights receptive-field group (a "plane" in neocognitron terminology) is often called a filter, and a layer typically has several such filters. "C-layer": a downsampling layer that contain units whose receptive fields cover patches of previous convolutional layers. Such a unit typically computes a weighted average of the activations of the units in its patch, and applies inhibition (divisive normalization) pooled from a somewhat larger patch and across different filters in a layer, and applies a saturating activation function. The patch weights are nonnegative

and are not trainable in the original neocognitron. The downsampling and competitive inhibition help to classify features and objects in visual scenes even when the objects are shifted. Several supervised and unsupervised learning algorithms have been proposed over the decades to train the weights of a neocognitron. Today, however, the CNN architecture is usually trained through backpropagation. Fukushima's ReLU activation function was not used in his neocognitron since all the weights were nonnegative; lateral inhibition was used instead. The rectifier has become a very popular activation function for CNNs and deep neural networks in general.